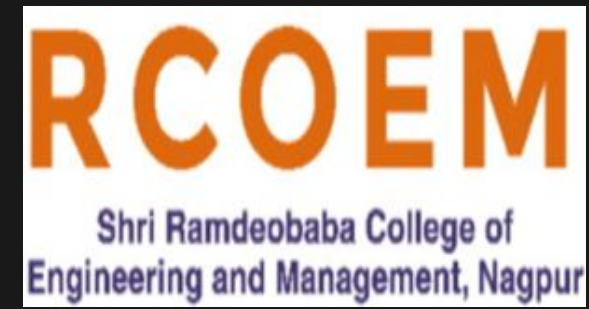




# **SHRI RAMDEOBABA COLLEGE OF ENGINEERING AND MANAGEMENT, NAGPUR**

## **Department of AI/CS**

**Programme : B.Tech Information Technology / B.Tech CSE (AI/CS)**

**Semester : VI**

## **PROJECT SEMINAR ON: AI-BASED INTERNSHIP RECOMMENDATION ENGINE**

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# The Challenges

- **The Participation Gap**

Despite **621,000 applications** for 127,000 internship opportunities in Phase 1, only **8,700 candidates** actually joined — a mere **10.6% conversion rate** .

- **Diverse User Base**

Youth from **rural areas, tribal districts, urban slums, and remote colleges** . Many are **first-generation learners** with limited digital exposure and no prior internship experience.

- **Decision Paralysis**

With **hundreds of internships** across 24 sectors, candidates struggle to identify which opportunities match their skills, interests, and aspirations, leading to misaligned applications.

## Phase 1 Statistics

**621K**

Applications Received

**127K**

Opportunities Posted

**8.7K**

Candidates Joined

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**10.6%**

Conversion Rate

# Target Users

## Who Are Our Actual Users?

- **Age 21-24** — Young adults seeking their first professional experience
- **Rural & Remote Areas** — Limited access to career guidance and mentorship
- **First-Generation Learners** — No family guidance on career pathways
- **Limited Digital Literacy** — Uncomfortable with complex digital interfaces
- **No Prior Experience** — First-time internship applicants unsure of expectations

### Geographic Distribution

735

Districts Covered

24

Sectors Available

# Current System Limitations

## ✔ What Works Today

- ✔ Online Portal
- ✔ Multi-Criteria Search
- ✔ 5 Applications
- ✔ Auto-Resume
- ✔ Mobile App

## Participation Stats

|                       |           |
|-----------------------|-----------|
| Total Companies       | 327       |
| Total Opportunities   | 1.18 Lakh |
| Applications Received | 4.55 Lakh |
| Offers Made           | 72,000+   |

## ✖ Critical Gaps

- 1No Intelligent Matching
- 2No Skill-Gap Analysis
- 3No Explainability
- 4Limited Accessibility

# Proposed Solution Overview



## AI-Based Internship Recommendation Engine

A **lightweight, mobile-first** solution that suggests **3-5 most relevant internships** to each candidate based on their profile, academic background, interests, and location preferences. Designed for users with limited digital literacy.



### Hybrid AI Engine

Combines content-based and collaborative filtering for accurate, personalized recommendations



### Mobile-First Design

Optimized for smartphones with touch-friendly interfaces and minimal text



### Regional Languages

Support for Hindi, Tamil, Telugu, Bengali and more for wider accessibility

### ⚙️ How It Works

- 1 **Simple Input:** 4-step wizard captures profile
- 2 **AI Analysis:** Hybrid engine processes data
- 3 **Smart Matching:** 3-5 top recommendations
- 4 **Clear Results:** Visual cards with explanations

### ★ Key Features

- ✓ Match percentage scoring
- ✓ "Why This Match?" explanations
- ✓ Visual skill alignment charts
- ✓ One-tap application integration
- ✓ Offline recommendation viewing
- ✓ Progress tracking dashboard

# AI Recommendation Engine

## Content-Based Filtering

Matches candidate profile with internship requirements using feature similarity.

-  **Education Matching**  
Compares degree, field of study, academic performance against requirements
-  **Skill Alignment**  
TF-IDF vectorization of skills with cosine similarity scoring
-  **Location Proximity**  
Geographic distance calculation with preference weighting
-  **Sector Interest**  
Matches candidate sector preferences with internship categories

## Collaborative Filtering

Learns from successful matches to recommend internships that similar candidates accepted.

- User Similarity Matrix**  
Identifies candidates with similar profiles, skills, and preferences
- Matrix Factorization**  
Lightweight SVD optimized for sparse interaction data
- Success Pattern Learning**  
Analyzes which internships were accepted by similar candidates

## Hybrid Scoring

- Content-Based Weight

60%
- Collaborative Weight

40%
- Dynamic Adjustment:

Weights shift based on data availability (more collaborative as user base grows)

# Technical Architecture

## Frontend

React 18

Tailwind CSS

## Backend

Python Flask API

RESTful APIs

## Data Layer

PostgreSQL

Redis Cache

Data Encryption

## AI/ML Layer

scikit-learn

Lightweight CF

## Deployment

### Containerization

Docker containers for consistent deployment

Docker

K8s

### Cloud Agnostic

Deploy on AWS, Azure

AWS

Azure

### Auto-Scaling

Horizontal scaling based on demand

### Integration

Seamless API integration with PM Internship portal

# User Interface Design

## Input Flow: 4-Step Wizard

- 1

**Education Level**  
10th/12th, ITI, Diploma, Graduate
- 2

**Skills Selection**  
Communication, Technical, Management, Creative
- 3

**Sector Interests**  
IT, Banking, Healthcare, Manufacturing, etc.
- 4

**Location Preference**  
Home state, nearby cities, pan-India

## Accessibility Features



44px Touch Targets



High Contrast



Regional Languages



Voice Support



Offline Mode



Easy Back/Reset

## Recommendation Cards



**Software Developer**  
**Intern**  
Tata Consultancy Services

**92%**  
Match

Mumbai

IT Sector

₹5,000/mo

### Why This Match?

- Your BCA degree aligns with requirements
- Programming skills match job description
- Mumbai is within your preferred location

View Details

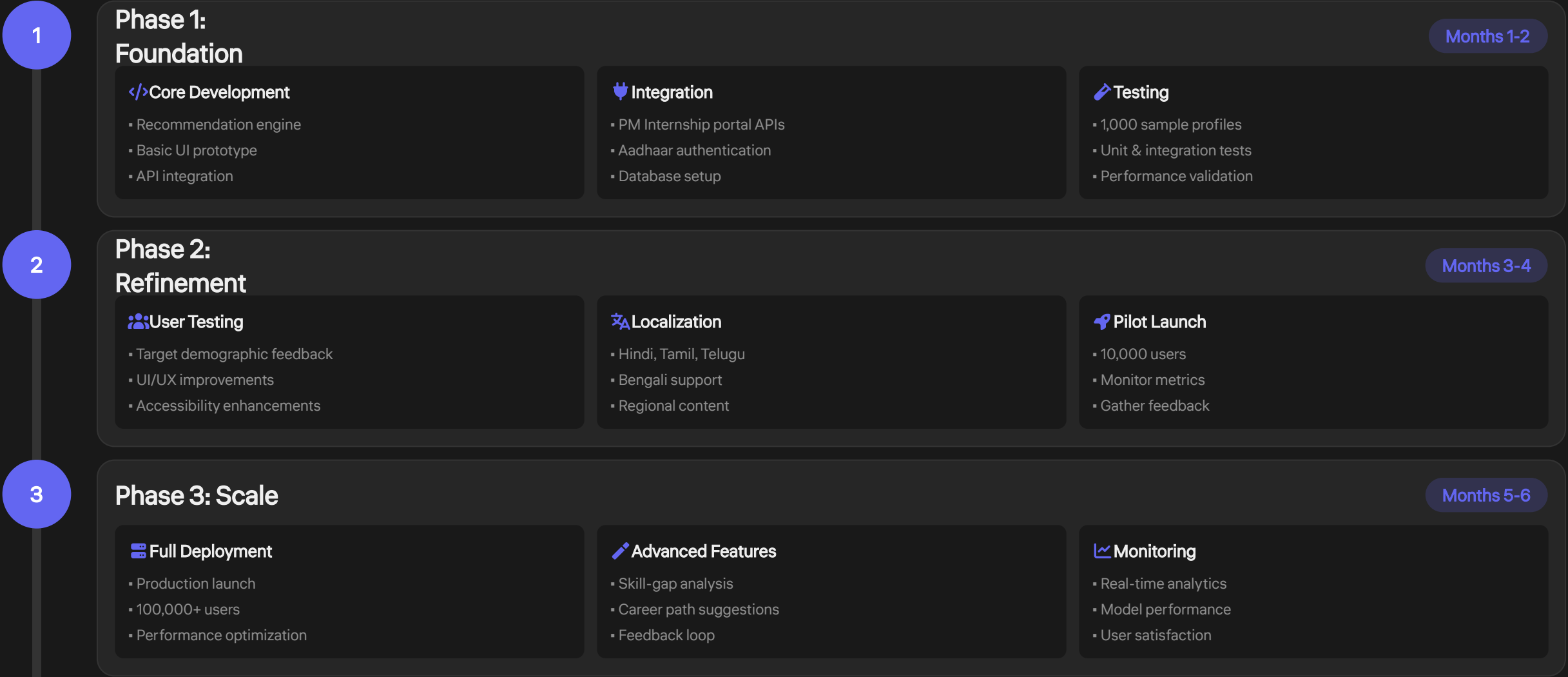


## Design Principles

- ✓ Minimal text, maximum visuals
- ✓ Progressive disclosure
- ✓ Consistent navigation patterns
- ✓ Clear error messages
- ✓ Immediate feedback on actions



# Implementation Roadmap



# Value Proposition & Expected Impact

|  For Candidates                                                                                                                                                                                                                |  For PMIS                                                                                                                                                                                                            |  For Companies                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>Time Saved70%<br/>Reduction in search time</div> <div>Match Quality3x<br/>Higher probability of suitable matches</div> <div>Confidence↑<br/>Informed decision-making</div> <div><div>✓ Better skill-job alignment</div><div>✓ Clear match explanations</div><div>✓ Reduced application anxiety</div></div> | <div>Conversion Rate35-40%<br/>Up from 10.6%</div> <div>Application Quality↑<br/>Better candidate-company fit</div> <div>Satisfaction↑<br/>Improved program experience</div> <div><div>✓ Reduced application overload</div><div>✓ Higher completion rates</div><div>✓ Data-driven insights</div></div> | <div>Screening Time-50%<br/>Faster candidate evaluation</div> <div>Candidate Quality↑<br/>Better matched applicants</div> <div>Retention↑<br/>Improved intern performance</div> <div><div>✓ Relevant applications only</div><div>✓ Reduced hiring costs</div><div>✓ Better ROI on CSR</div></div> |

## Key Success Metrics

85%+

Match Accuracy

4.5/5

User Satisfaction

3x

Conversion Rate

<2s

Response Time

# Future Enhancements

## AI-Powered Career Guidance

### Skill-Gap Analysis

Identify missing skills for target roles and recommend learning paths

NSDC Integration

Online Courses

### Career Path Visualization

Show progression from internship → full-time → leadership roles

Timeline View

Skill Milestones

### Mentorship Matching

Connect candidates with industry mentors based on career goals

Video Calls

Chat Support

## Advanced Features

### Chatbot Assistant

AI-powered chatbot for queries in regional languages with voice support

### Video Introductions

Candidates can record short video intros to complement applications

### Culture Matching

Match candidates with companies based on work culture preferences

### Salary Insights

Show stipend comparisons and post-internship salary trends

## Enhanced Personalization

### Behavioral Learning

Learn from views, saves, and applications

### Seasonal Trends

Analyze internship availability patterns

### Success Prediction

Predict application success probability

## Integration Expansion

 **Skill India:** Connect with training programs

 **Universities:** Credit recognition system

 **Job Portals:** Post-internship placements

 **Certifications:** Digital credential wallet

## CONCLUSION

# Key Takeaways



### Measurable Impact

Expected outcomes: **3x improvement** in conversion rates, significantly better candidate-company matches, and enhanced user satisfaction.



### Balanced AI Approach

Hybrid recommendation engine combines **accuracy with explainability**, building user trust through transparent "Why This Match?" explanations.



### Bridging the Digital Divide

This solution demonstrates how thoughtful AI implementation can create meaningful opportunities for underserved communities and contribute to **Viksit Bharat** vision.

**Thank you**